Robust Heart-Rate Estimation from Phone-Camera rPPG: Preprocessing, Channel Selection, Fusion, and Tracking

1 Signal model

Let $c \in \{R, G, B\}$ index the camera color channels and $x_c(t)$ be the mean (or spatially pooled) intensity in a skin ROI at time t. A standard small-signal rPPG model is

$$x_c(t) = \underbrace{L_c(t)}_{\text{slow lighting}} \big(1 + \alpha_c\, p(t)\big) + \eta_c(t),$$

where $L_c(t)$ captures slow illumination/pose changes, $p(t)$ is the pulsatile waveform (shared across channels), α_c are channel gains (skin optics), and η_c is noise (read noise, motion, quantization).

Define a normalized channel

$$y_c(t) = \frac{x_c(t)}{\overline{x_c(t)}} - 1,$$

where $\overline{x_c(t)}$ is a local mean (e.g., moving average or polynomial trend). Then

$$y_c(t) \approx \alpha_c\, p(t) + \epsilon_c(t), \qquad \epsilon_c(t) \text{ small.}$$

2 Preprocessing

(P1) **ROI pooling:** spatially average skin pixels to reduce sensor noise:

$$x_c(t) = \frac{1}{N} \sum_{i \in \mathrm{ROI}} I_{c,i}(t).$$

(P2) **Detrending / illumination correction:** remove $L_c(t)$ by high-pass or polynomial detrending. For sampling rate f_s, use a zero-phase Butterworth HP at $\sim 0.3\,\mathrm{Hz}$:

$$\tilde{x}_c(t) = \mathrm{HP}_{0.3\ \mathrm{Hz}}\{x_c(t)\}.$$

Or normalize: $y_c(t) = x_c(t)/\mathrm{MA}_\tau(x_c(t)) - 1$ with window $\tau \in [1\,\mathrm{s}, 5\,\mathrm{s}]$.

(P3) **Band-pass:** keep physiological band $[f_{\min}, f_{\max}]$, e.g., $[0.7, 4.0]$ Hz ($42 - -240\,\mathrm{bpm}$):

$$z_c(t) = \mathrm{BP}_{[f_{\min}, f_{\max}]}\{y_c(t)\}.$$

(P4) **Temporal windowing:** use sliding windows of length T seconds with overlap ρ (e.g., $T = 8\,\mathrm{s}$, $\rho = 50\%$). All subsequent estimates are per-window k.

3 Channel quality and selection

Per window k, compute the (Welch) periodogram $P_c^{(k)}(f)$ of z_c. Let

$$f_{\text{peak},c}^{(k)} = \arg\max_{f\in[f_{\min},f_{\max}]} P_c^{(k)}(f), \qquad \text{SNR}_c^{(k)} = \frac{\int_{\mathcal{B}(f_{\text{peak},c}^{(k)})} P_c^{(k)}(f)\,df}{\int_{\mathcal{N}} P_c^{(k)}(f)\,df},$$

where $\mathcal{B}$ is a narrow band around the peak (e.g., ± 0.1 Hz) and $\mathcal{N}$ excludes the peak band. A simple *winner-takes-all* selector is

$$c_k^\star = \arg\max_{c\in\{R,G,B\}} \text{SNR}_c^{(k)}.$$

This directly adapts to which channel is strongest (red/green/blue variability).

4 Chrominance-based fusion (color space methods)

Let r, g, b be the normalized, band-passed channels z_R, z_G, z_B.

CHROM

Define

$$X = 3r - 2g, \qquad Y = 1.5r + g - 1.5b.$$

Variance-normalized fusion:

$$s_{\text{CHROM}}(t) = X(t) - \frac{\sigma_X}{\sigma_Y} Y(t),$$

where $\sigma_\cdot$ are standard deviations computed in window k. Heart rate is obtained from s_{CHROM} as in Sec. 7.

POS

Define

$$S = r - g, \qquad H = r + g - 2b,$$

and

$$s_{\text{POS}}(t) = S(t) - \frac{\sigma_S}{\sigma_H} H(t).$$

5 Blind-source fusion (ICA/PCA)

Stack signals $z(t) = [z_R(t), z_G(t), z_B(t)]^\top$ in window k, whiten, and estimate an unmixing matrix W:

$$s(t) = W\,z(t), \qquad s \in \mathbb{R}^3.$$

Pick the component

$$j^\star = \arg\max_{j\in\{1,2,3\}} \max_{f\in[f_{\min},f_{\max}]} P_{s_j}^{(k)}(f).$$

Use $s_{j^\star}(t)$ for HR estimation.

6 Adaptive spectral weighting

Compute weights $w_c^{(k)} \geq 0$ with $\sum_c w_c^{(k)} = 1$, e.g.,

$$w_c^{(k)} = \frac{\mathrm{SNR}_c^{(k)}}{\sum_{c'} \mathrm{SNR}_{c'}^{(k)}}.$$

Fuse

$$s_{\mathrm{w}}(t) = \sum_{c \in \{R,G,B\}} w_c^{(k)}\, z_c(t).$$

7 Heart-rate estimators

Let $s(t)$ be any of the signals produced above in the current window.

Frequency-domain (periodogram/Welch)

Compute $P^{(k)}(f)$. Then

$$f_k^\star = \arg \max_{f \in [f_{\min}, f_{\max}]} P^{(k)}(f), \qquad \widehat{\mathrm{HR}}_k = 60\, f_k^\star \text{ bpm}.$$

A sub-bin refinement uses quadratic interpolation around the discrete peak.

Autocorrelation

Compute the unbiased autocorrelation

$$R(\tau) = \int s(t)s(t+\tau)\, dt,$$

or its discrete equivalent. Restrict lags $\tau \in [\tau_{\min}, \tau_{\max}]$ with $\tau_{\min} = 1/f_{\max}$, $\tau_{\max} = 1/f_{\min}$. Then

$$\tau_k^\star = \arg \max_{\tau \in [\tau_{\min}, \tau_{\max}]} R(\tau), \qquad \widehat{\mathrm{HR}}_k = \frac{60}{\tau_k^\star} \text{ bpm}.$$

Peak-picking on time domain

Detect systolic peaks $\{t_i\}$ in $s(t)$. Let inter-beat intervals $\Delta_i = t_{i+1} - t_i$. A robust estimate is

$$\widehat{\mathrm{HR}}_k = \frac{60}{\mathrm{median}_i\, \Delta_i} \text{ bpm},$$

with refractory period and prominence thresholds to avoid false peaks.

Lomb–Scargle (irregular sampling)

For uneven timestamping, maximize the Lomb–Scargle periodogram $P_{\mathrm{LS}}(f)$:

$$f_k^\star = \arg \max_{f \in [f_{\min}, f_{\max}]} P_{\mathrm{LS}}(f), \qquad \widehat{\mathrm{HR}}_k = 60 f_k^\star.$$

Time–frequency ridge (CWT)

Compute the continuous wavelet transform $W_s(a,b)$ (Morlet). Map scales a to frequencies f. Follow the energy ridge $f^\star(b)$ maximizing $|W_s(a,b)|$ and take $\widehat{\mathrm{HR}}_k = 60\, \mathrm{mode}_b f^\star(b)$.

Parametric spectral (AR)

Fit an AR(p) model to $s(t)$ and use its PSD $P_{\mathrm{AR}}(f)$ to locate the dominant $f_k^\star$.

8 Temporal tracking and outlier control

Let h_k be the latent heart rate (Hz) per window k. Enforce physiological smoothness.

Kalman-style tracking

State model:

$$h_k = h_{k-1} + \nu_k, \qquad \nu_k \sim \mathcal{N}(0, q).$$

Measurement from spectrum:

$$y_k = f_k^\star + \epsilon_k, \qquad \epsilon_k \sim \mathcal{N}(0, r_k).$$

Run a (scalar) Kalman filter on h_k. Set r_k inversely to spectral sharpness/SNR:

$$r_k \propto \frac{\text{noise floor}}{\text{peak power}}.$$

Return $\widehat{\mathrm{HR}}_k = 60\,\hat{h}_k$.

Dynamic-programming frequency path

On a frequency grid $\{f_m\}$, define window costs

$$C_k(f_m) = -\log P^{(k)}(f_m),$$

and a smoothness penalty $\lambda(f_m - f_{m'})^2$. Minimize

$$\{f_k^\star\} = \arg\min_{f_1,\ldots,f_K} \sum_k C_k(f_k) + \lambda \sum_k (f_k - f_{k-1})^2,$$

via Viterbi. Convert $f_k^\star$ to bpm.

9 Quality metrics and confidence

Define a per-window confidence Γ_k combining spectral concentration and inter-method agreement:

$$\Gamma_k = \underbrace{\frac{P^{(k)}(f_k^\star)}{\int_{f_{\min}}^{f_{\max}} P^{(k)}(f)\,df}}_{\text{spectral concentration}} \times \underbrace{\exp\left(-\frac{(f_k^\star - f_k^{\mathrm{AC}})^2}{2\sigma^2}\right)}_{\text{agreement with autocorr}},$$

where f_k^{AC} is the autocorrelation-based frequency. Use Γ_k to gate outputs and adjust r_k in the tracker.

10 Motion-robustness add-ons

(M1) **Frame-quality gating:** discard frames with ROI blur, low brightness, or heavy compression artifacts (simple thresholds on variance and mean).

(M2) **Color-ratio stabilization:** work with ratios x_c/x_{luma} to reduce multiplicative lighting.

(M3) **Multi-ROI consensus:** compute per-ROI HR and take a robust average or the mode in frequency.

11 End-to-end recipe (practical default)

Per window k:

1. ROI pool $\to x_c(t)$; detrend/normalize; band-pass to $z_c(t)$.
2. Compute $P_c^{(k)}(f)$ and $\mathrm{SNR}_c^{(k)}$.
3. Build three candidate signals: (i) best channel $c_k^\star$, (ii) CHROM/POS $s_{\mathrm{CHROM/POS}}$, (iii) weighted fusion s_{w}.
4. For each candidate, estimate HR via periodogram peak with sub-bin interpolation; cross-check with autocorrelation and (if irregular) Lomb–Scargle.
5. Pick the frequency maximizing confidence Γ_k; feed to a Kalman tracker with $r_k \propto 1/\mathrm{SNR}$.
6. Output $\widehat{\mathrm{HR}}_k$ with a confidence score and suppress jumps $> 15\,\mathrm{bpm}$ unless confidence is high.

12 Parameter suggestions (typical phone setup)

Sampling $f_s \in [30\,\mathrm{Hz}, 60\,\mathrm{Hz}]$; window $T = 8\,\mathrm{s}$, overlap 50%; band-pass $[0.7, 4.0]$ Hz; Welch segments: 4 with 50% overlap; peak band $\pm 0.1\,\mathrm{Hz}$ for SNR; tracker: q tuned for $3\,\mathrm{bpm/s}$ max acceleration, r_k from spectral sharpness.

13 Uncertainty estimate

Approximate bpm uncertainty from peak curvature. If a quadratic fit around $f^\star$ yields $P(f) \approx a(f - f^\star)^2 + b$, then

$$\sigma_f \approx \sqrt{\frac{\sigma_n^2}{-2a}}, \qquad \sigma_{\mathrm{bpm}} \approx 60\,\sigma_f,$$

with σ_n^2 an estimate of spectral noise floor in the band.